

CANDIDATE'S NAME: _____ Group: _____

YISHUN JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION 2018

CHEMISTRY HIGHER 1 Paper 2

8873/02
27 August 2018
0800hrs – 1000hrs
2 hours

Additional materials:
Data Booklet
Electronic calculator

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INSTRUCTIONS TO CANDIDATES

Write your name and group in the spaces at the top of this page.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use paper clips, highlighters, glue or correction fluid.
The use of scientific calculator is expected, where appropriate.

Answer **all** the questions on the Question Paper.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

Paper 1

Total	/ 30
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Paper 2

1	/ 28
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2	/ 12
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3	/ 8
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4	/ 6
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5	/ 6
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6	/ 20
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7	/ 20
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Total	/ 80
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Overall	/110
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Section A

Answer **all** the questions in this section in the spaces provided.

1. The Earth's climate is changing. Temperatures are rising, snow and rainfall patterns are shifting, and more extreme climate events – like heavy rainstorms and record high temperatures – are already happening. Many of these observed changes are linked to the rising levels of carbon dioxide and other greenhouse gases (gases that trap heat) in our atmosphere, caused by human activities.

Gross calculation shows that while carbon dioxide accounts for 74% of the greenhouse gases, methane is the second most prevalent (14%) greenhouse gas emitted from human activities - natural gas (mainly methane) and petroleum systems activities such as incinerations, are the largest source of methane emission, accounting for 31%, while methane generated from landfills as waste decomposes, accounting for 16%.

Although methane's lifetime in the atmosphere is much shorter than carbon dioxide, but methane is more efficient at trapping radiation than carbon dioxide. Kilogram for kilogram, the comparative impact of methane is about 12 times greater than carbon dioxide.

A fossil fuel power station is a power station which burns a fossil fuel such as coal, natural gas, or petroleum to produce electricity. In an attempt to cut down carbon dioxide emissions during the production of electricity from such fuels, there has been a move in many countries to build more power stations burning natural gas rather than coal. The following table compares these two types of power station.

Type of power station	Overall efficiency of power station	Amount of by-product produced per MJ of electrical energy (1 MJ = 10^6 J)	
		SO ₂	NO ₂
coal	40%	0.31 g	0.64 g
natural gas	51%	0.0015 g	0.11 g

In your calculations, assume that coal consists of 95% of carbon, together with 5% of non-combustible ash.

(a) Write balanced equations for the complete combustion of

(i) methane

..... [1]

(ii) carbon

..... [1]

(b) The enthalpy changes of combustion of methane and carbon are :

$$\Delta H_c(\text{CH}_4) = -890 \text{ kJ mol}^{-1}$$

$$\Delta H_c(\text{C}) = -394 \text{ kJ mol}^{-1}$$

- (i) Use these data to calculate how many moles of each of methane and carbon needed to be burned to produce 1 MJ of **heat** energy. [2]

Methane :

Carbon :

- (ii) Using the efficiency figures given in the table, and your answers in (b)(i), calculate how many moles of each of methane and carbon needed to be burned in the power stations, to produce 1 MJ of **electrical** energy. [4]

Methane :

Carbon :

- (c) Hence, calculate the mass of the ash that would be produced per MJ of electrical energy in a coal-fired power station. Assume that the average relative molecular mass of the ash is 220.0. [2]

[Turn over

- (d) Use the data in the table to suggest **two** environmental advantages of using a natural gas power station. Explain your answer. [3]

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- (e) Despite the advantages of operating a gas power station, use your answer in **b(ii)** and relevant data from the passage, explain why methane is more harmful to the environment than coal. [3]

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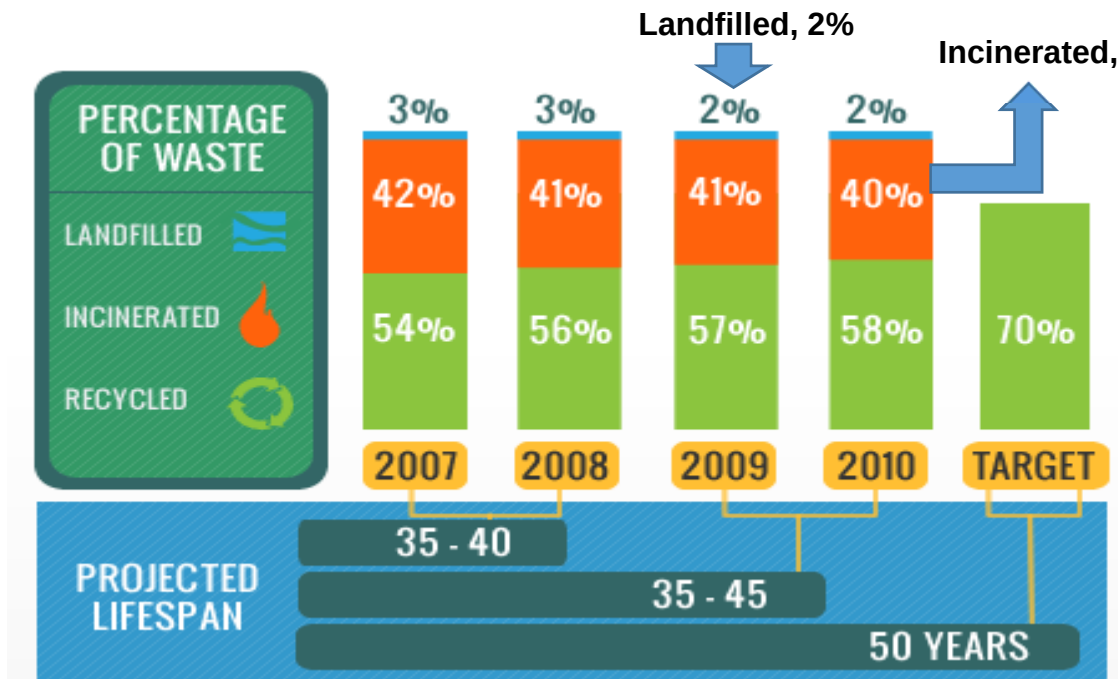
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- (f) Locally, the National Environment Agency, NEA has released the following data on its refuse disposal :



Use relevant data from the passage, suggest one advantage and one disadvantage for NEA to increase the use of incineration for refuse disposal over landfill. [2]

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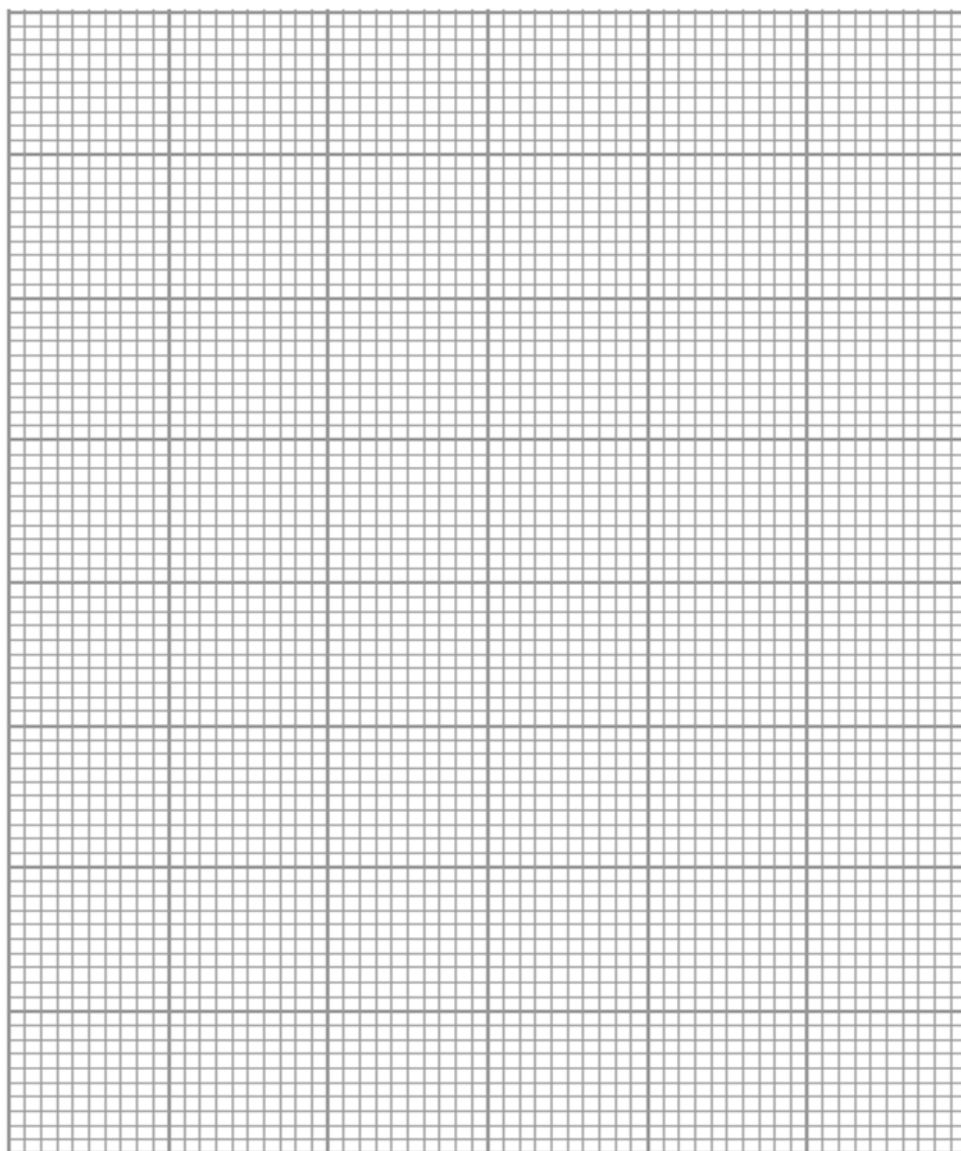
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[Turn over

- (g) Methane belongs to a family of organic compounds known as alkanes. The standard enthalpy changes of combustion of five straight chain alkanes are given in the table below.

alkane	CH ₄	C ₂ H ₆	C ₃ H ₈	C ₄ H ₁₀	C ₅ H ₁₂
$\Delta H_c^\ominus / \text{kJ mol}^{-1}$	–890	–1560	–2230	–2920	–3570

On the grid below, plot a graph to show how the standard enthalpy changes of combustion vary for these five alkanes. [3]



Use the graph you have plotted above, determine

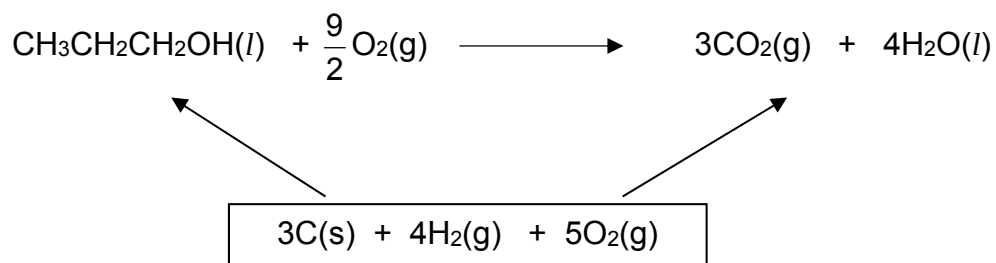
(i) the **average** standard enthalpy change of combustion of a "CH₂" group [2]

(ii) the standard enthalpy change of combustion of hydrogen gas [1]

(iii) the standard enthalpy change of combustion of hexane, C₆H₁₄. [2]

(h) Alcohols are also commonly used as fuels.

The diagram below shows an energy cycle involving propan-1-ol, CH₃CH₂CH₂OH.



Use the following data to calculate the standard enthalpy change of combustion of propan-1-ol. [2]

$$\Delta H_f^\ominus(\text{CO}_2) = -393 \text{ kJ mol}^{-1}$$

$$\Delta H_f^\ominus(\text{H}_2\text{O}) = -286 \text{ kJ mol}^{-1}$$

$$\Delta H_f^\ominus(\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}) = -303 \text{ kJ mol}^{-1}$$

[Total : 28]

[Turn over]

2. The mass spectrometer made it possible to determine the masses of atoms very accurately, and it became clear that not all atoms of the same element had the same mass.

The accurate masses gave rise to the idea of the isotopes:

“Isotopes have the same atomic numbers, but different mass numbers.”

- (a) (i) Write the symbols showing the nucleon numbers, proton numbers and charges of the following three particles, **K**, **L** and **M**.

You may refer to the example given for the particle, **A**

[3]

Particle	Protons	Neutrons	Electrons	Symbol
K	6	8	6	
L	7	7	10	
M	8	7	7	
A	5	8	5	$^{13}_{5}\text{A}$

- (ii) Using the data from the *Data Booklet*, write the symbols of the usual isotopes of the elements concerned.

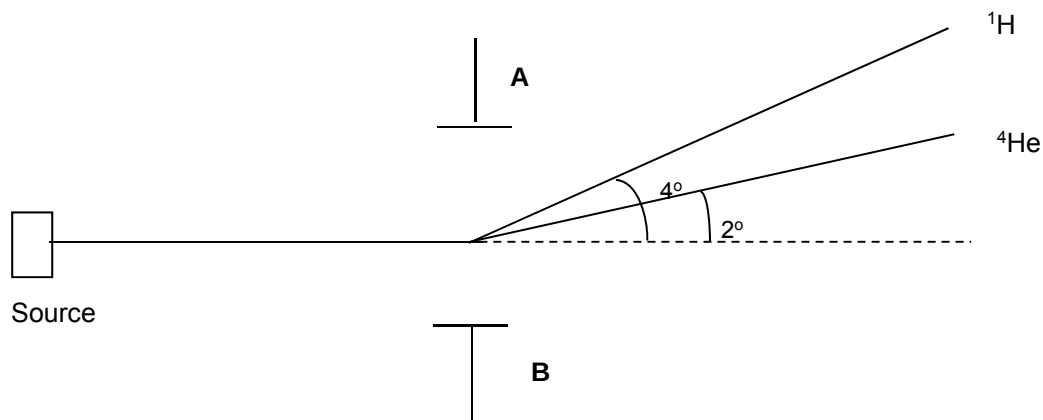
[3]

Particle **K** :

Particle **L** :

Particle **M** :

- (b) A plasma is a gaseous mixture in which the atoms have been completely stripped of their electrons, leaving bare nuclei. Because of possible use in controlled nuclear fusion reactions, plasma behaviour has been extensively studied. When passed between two plates carrying a certain electric charge, ^1H and ^4He nuclei are deflected as follows.



Giving reason for your answer, suggest the polarity (+ or -) of plate A.

[2]

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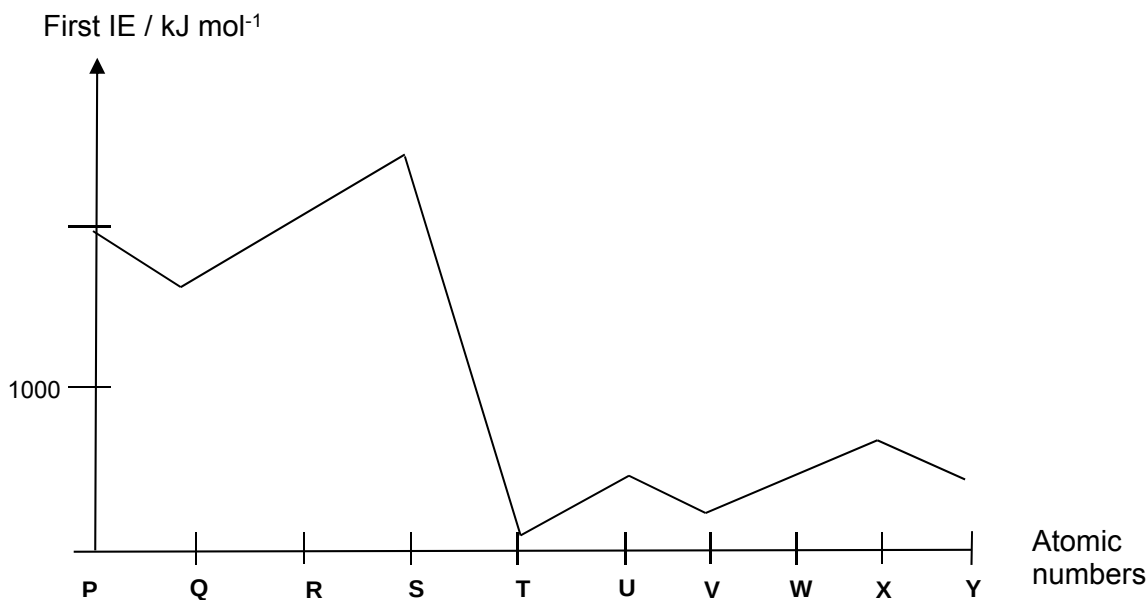
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[Turn over

- (c) In an analytical chemistry laboratory, the plasma torch is often used to ionise a variety of samples during their analyses.

In one series of analyses, samples of unknown elements **P** to **Y** were passed through the plasma torch and their first ionisation energies were recorded and plotted as shown.



P to **Y** are consecutive elements with atomic numbers below 20.

- (i) Identify the group to which element **X** belongs to, and state its electronic configuration. [2]

Group :

Electronic configuration :

- (ii) Give the formula of the chloride of **X**, able to function as a Lewis base, and draw the dot-and-cross diagram of this chloride. [2]

Chloride of **X** :

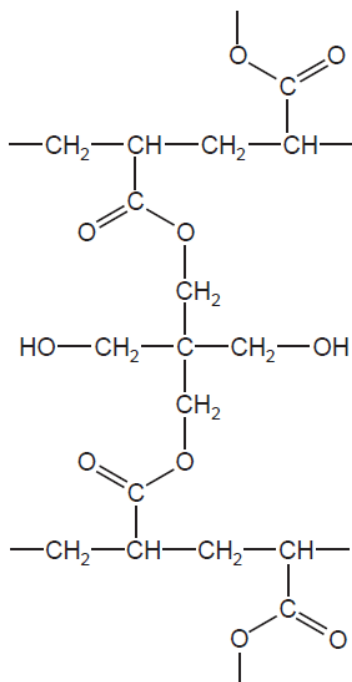
Dot-and-cross diagram of the chloride of **X**



[Total : 12]

3. In recent years there has been considerable interest in a range of polymers known as 'hydrogels'. These polymers are hydrophilic and can absorb large quantities of water.

(a) The diagram shows part of the structure of a hydrogel.



The hydrogel is formed from the chains of one polymer which are cross-linked using another molecule.

- (i) Draw the structure of a possible monomer used in the polymer chains. [1]

- (ii) State the type of polymerisation used to form these chains. [1]

.....

- (iii) Draw the structure of the molecule used to cross-link the polymer chains. [1]

[Turn over

- (b) Once a hydrogel has absorbed water, it can be dried and re-used many times.

Explain why this is possible, making reference to the structure shown on the previous page.

[2]

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- (c) Not every available side-chain in the polymer is cross-linked, and the amount of cross-linking affects the properties of the hydrogel.

- (i) The amount of cross-linking has little effect on the ability of the gel to absorb water.

Suggest why this is the case.

[1]

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- (ii) Suggest **one** property of the hydrogel that will change if more cross-linking takes place.

Explain how the increased cross-linking brings about this change.

[2]

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[Total: 8]

4. In search of the perfect hair dye, materials scientists conducted experiments whereby they coated hair with a graphene solution that included water, vitamin C and a polymer to improve adhesion. They reported that the new method stayed on after 30 washings, the number necessary for a hair dye to be considered permanent.

(a) State the difference between a nanomaterial and nanoparticle in terms of size. [1]

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(b) Graphene is a nanomaterial which is strong, thin and a good conductor of electricity.

- (i) Describe the structure of graphene. Explain how this structure gives graphene the properties of high strength and high electrical conductivity.

You may find it useful to include a labelled diagram in your answer. [3]

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[Turn over

- (ii) Suggest why a solution of graphene makes a good coating material for hair. [2]

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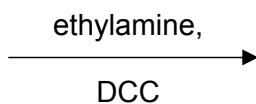
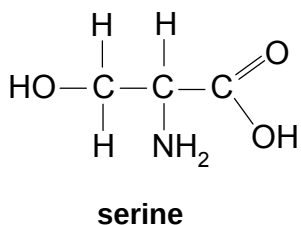
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[Total: 6]

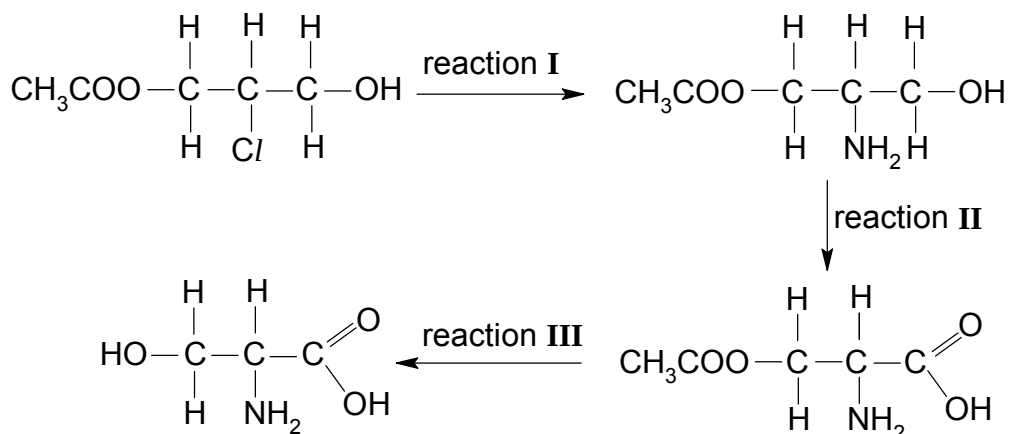
5. Amino acids play central roles both as building blocks of protein and as intermediates in metabolism.

(a) Serine is a naturally-occurring amino acid.

- (i) In the box below, draw the structure of the organic compound formed when serine reacts with ethylamine in the presence of dicyclohexylcarbodiimide, DCC. [1]



One possible method of synthesising serine is shown below.



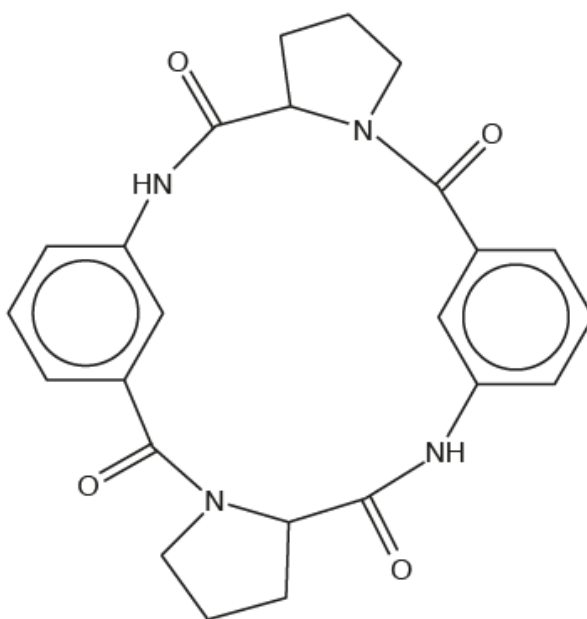
- (ii) State the types of reaction for reactions I and II. [2]

reaction I:

reaction II:

[Turn over

(b) A cyclic tetrapeptide has been synthesised from 3-aminobenzoic acid and an amino acid.



cyclic tetrapeptide

The cyclic tetrapeptide was treated with hot aqueous sodium hydroxide.

Draw the structures of the **two** organic products formed from the reaction and state the type of reaction undergone. [3]

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type of reaction:

[Total:6]

Section B

Answer **one** question from this section on separate answer paper.

6. (a) Carboxylic acids are important organic compounds which are widely used as food flavourings. Carboxylic acids are usually *weak Brønsted acids*.

- (i) Explain what is meant by a *weak Brønsted acid*. [2]

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- (ii) The K_a values for some organic acids are listed below.

acid	$K_a / \text{mol dm}^{-3}$
$\text{CH}_3\text{CO}_2\text{H}$	1.7×10^{-5}
$\text{ClCH}_2\text{CO}_2\text{H}$	1.3×10^{-3}
$\text{Cl}_2\text{CHCO}_2\text{H}$	5.0×10^{-2}

- Explain the trend in K_a values in terms of the structure of these acids. [3]

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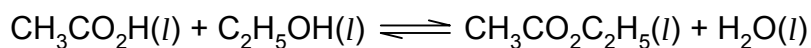
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[Turn over

- (b) Esters such as ethyl ethanoate can be formed by reacting carboxylic acids with alcohols.



This reaction is allowed to proceed until a dynamic equilibrium is established.

- (i) Explain what is meant by the term *dynamic equilibrium*. [1]

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- (ii) Write the expression for the equilibrium constant, K_c , for this reaction. [1]

- (iii) For this equilibrium, the value of K_c is 4.0 at 298 K.

A mixture containing 0.5 mol ethanoic acid, 0.5 mol ethanol, 0.1 mol ethyl ethanoate and 0.1 mol water was set up and allowed to come to equilibrium at 298 K.

Calculate the amount, in moles, of each substance present at equilibrium. [3]

- (iv) State and explain what would happen to the position of the equilibrium when 0.05 mol ethyl ethanoate is withdrawn from the equilibrium mixture in (b)(iii). [2]

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- (c) **H** is a halogenoalkane with molecular formula C_4H_9Cl . When reacted with hot ethanolic NaOH, the product **J** is found to display cis-trans isomerism. When refluxed with aqueous NaOH, compound **K** is formed.

K undergoes oxidation to form product **L**. **L** does not react with Tollens' reagent.

- (i) Using the information given above, suggest the structures of compounds **H**, **J**, **K** and **L**.
[4]

- (ii) Draw and name the two isomers of compound **J**.
[2]

- (iii) Compound **K** is found to have a higher boiling point as compared to compound **L**. With reference to structure and bonding, explain why this is so.
[2]

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[Total : 20]

[Turn over

7. Phosphorus, P, is an element found in the third period of the Periodic Table. The element was discovered by Hennig Brand when he experimented with urine in 1669.

(a) (i) Sketch a graph showing the variation of melting point of the elements across the third period of the Periodic Table. [2]

(ii) Explain qualitatively the shape of the graph you have drawn in (a)(i). [2]

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Magnesium and aluminium are also elements from the third period of the Periodic Table. Both elements react with excess oxygen gas to form their respective oxides, MgO and Al_2O_3 .

At room temperature, both MgO and Al_2O_3 are white crystalline solids. These compounds have useful applications in our daily lives – MgO is used as a drying agent to preserve books and Al_2O_3 is used as an abrasive in sandpapers.

(b) A sample of white powder is known to be either MgO or Al_2O_3 . In order to determine its identity, **one** of the following reagents can be used.

- dilute hydrochloric acid
- aqueous sodium hydroxide
- aqueous hydrogen peroxide

(i) State the reagent you would use to determine the identity of the white powder. [1]

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(ii) Explain your choice in **(b)(i)**, and write balanced equation(s) for any chemical reactions that may occur. [3]

Equation(s) :

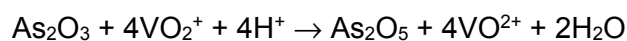
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Explanation :

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[Turn over

Arsenic, As, is in the same Group of the Periodic Table as phosphorus. The element burns in excess oxygen gas to form arsenic(III) oxide, As_2O_3 . An acidified solution of vanadium(V) ions is able to gradually oxidise arsenic(III) oxide to arsenic(V) oxide, according to the following balanced equation.

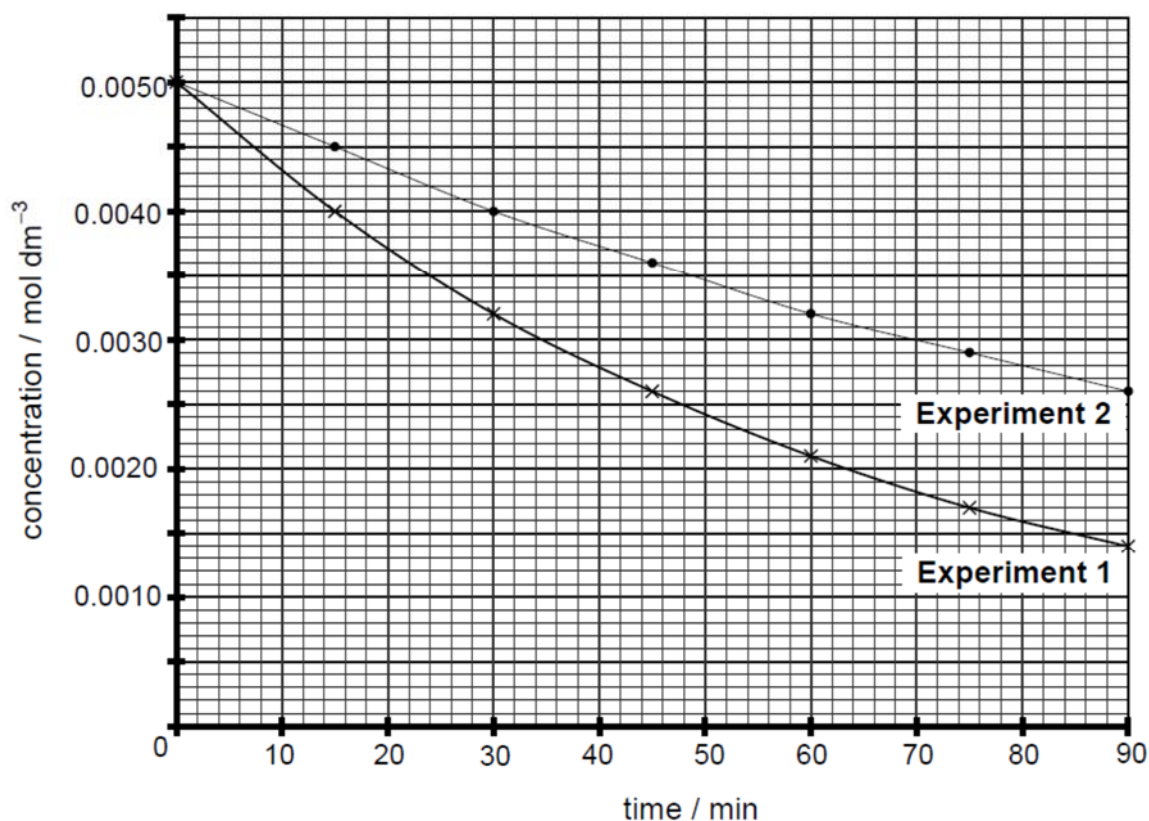


- (c) The rate of the reaction between acidified VO_2^+ ions and arsenic(III) oxide at 25°C was followed by measuring the concentration of remaining vanadium(V) ion after fixed time intervals. Two experiments were carried out, starting with different concentrations of arsenic(III) oxide.

Experiment 1: $[\text{As}_2\text{O}_3] = 0.10 \text{ mol dm}^{-3}$

Experiment 2: $[\text{As}_2\text{O}_3] = 0.05 \text{ mol dm}^{-3}$

The following graphs show the results of the two experiments.



- (i) Define *order of reaction*.

[1]

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- (ii) Use the graphs to determine the order of reaction with respect to arsenic (III) oxide, As_2O_3 and vanadium (V) oxide ion, VO_2^+ respectively. [4]

- (iii) Write the rate equation for the reaction. [1]

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- (iv) Calculate the rate constant for the reaction, stating its units. [2]

[Turn over

- (v) A third experiment involving the reaction between As_2O_3 and VO_2^+ is carried out at 40°C .

With the aid of a suitable diagram, state and explain how the rate of reaction at 40°C would compare with that at 25°C . [4]

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[Total : 20]

END OF PAPER